sixtytwo

Reliability infrastructure for AI pretraining

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OUR THESIS

A growing number of companies are pretraining their own models.

Data control and GPU economics are pushing companies on-prem. But going on-prem means taking on everything a cloud provider does invisibly — with none of their tooling.

THE PROBLEM

Pretraining runs are fragile

A single node failure days into a multi-week run can mean lost compute, manual debugging, and restarted training from stale checkpoints.

419 failures in 54 days on Meta's 16,384-GPU Llama 3 cluster

1 / 3hrs failure frequency at that scale

~9% annualized GPU failure rate

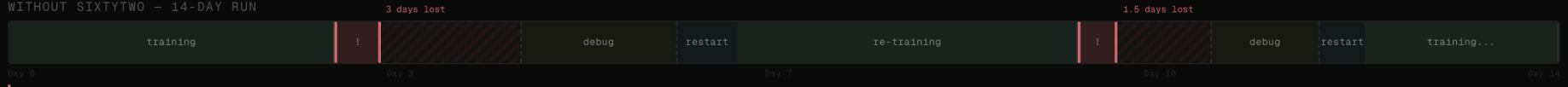
Source: Meta HPCA 2025, arXiv 2410.21680

```
03:14:22  node-047 step 48,291 loss: 2.847
03:14:23  node-047 step 48,292 loss: 2.843
03:14:24  node-012 NCCL TIMEOUT — allreduce hang
03:14:24  node-012 GPU 3: Xid 79 — fallen off bus
03:14:25  ALERT training stalled — rank 12 unresponsive
03:14:25  all ranks NCCL watchdog timeout — aborting
03:14:26  FATAL run terminated — 72h compute lost
07:22:00  eng-oncall paged — investigation begins
09:41:00  root cause: GPU HBM3 ECC error
11:30:00  node replaced, run restarted from scratch
```

PRETRAINING DEEP DIVE

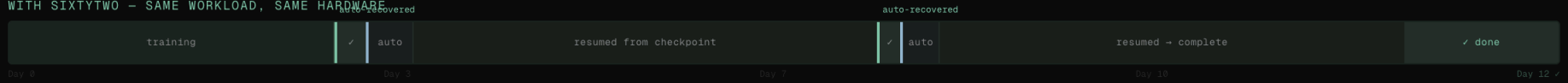
Anatomy of a failed training run

WITHOUT SIXTYTWO — 14-DAY RUN



~30% compute wasted. Run still incomplete at day 14.

WITH SIXTYTWO — SAME WORKLOAD, SAME HARDWARE



<5% overhead. Failures auto-detected, checkpointed, recovered. Finished 2 days early.

Compute utilization: 95%+ vs ~65% without

OUR SOLUTION

Self-healing training infrastructure

Sixtytwo monitors every layer of the training stack — hardware, network, framework — and automatically recovers from failures. No pager. No manual restarts.

01 **Detect**
Continuous health monitoring across GPU, NIC, NVLink, NCCL. Sub-second anomaly detection.

02 **Checkpoint**
Automatically saves training state when issues are detected. No progress lost.

03 **Replace**
Quarantines failing node, provisions a healthy spare. Handles topology reconfiguration.

04 **Resume**
Restores from checkpoint on new topology. Recovery in minutes, not days.

WHAT WE BUILD

The on-prem training platform

Observability

Unified monitoring across GPU health, training metrics, network fabric, and thermals. Purpose-built for pretraining.

Prometheus | DCGM | OpenTelemetry

Fleet Management

Inventory tracking, health scoring, automatic quarantine. Know which nodes are reliable and which are degrading.

Node health | Spare pool | Topology

Orchestration

Intelligent scheduling with automatic checkpointing, failure recovery, and run resumption for distributed training.

PyTorch | Kubernetes | Slurm

Container & Image Mgmt

Reproducible training environments with version-controlled containers. Fast distribution across hundreds of nodes.

Docker | NCCL | CUDA

WHY NOW

The shift to on-prem is accelerating

79% of enterprises have already repatriated some AI workloads from public cloud. The infrastructure to operate those clusters reliably doesn't exist yet.

Source: Feb 2026 enterprise IT survey (n=203)

01 **Data sovereignty**
73% of enterprises cite data privacy as top AI risk. 120+ countries now have data protection laws, up from 76 in 2011.

02 **Cost at scale**
On-prem breaks even vs cloud in under 4 months at high utilization. 8–18x cheaper per token for sustained workloads.

03 **Regulation**
EU AI Act high-risk rules take effect Aug 2026. Emerging US and global policy requires data residency and auditability.

04 **Open-source models**
Llama, Mistral, DeepSeek mean companies can pretrain on their own data. The workloads are here — the tooling isn't.

MARKET

A massive, growing opportunity

$758B

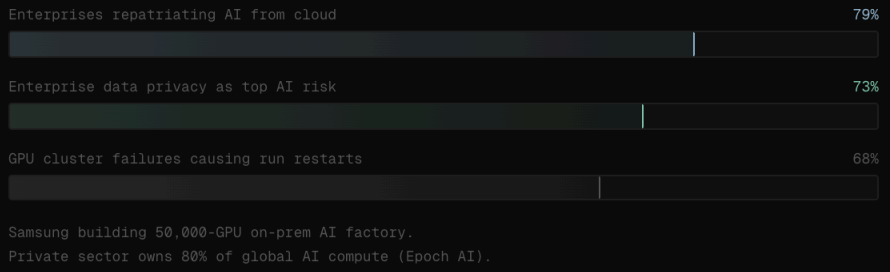
AI infrastructure spending by 2029 (IDC)

75%

of enterprise AI workloads on hybrid infra by 2028 (IDC)

$700M

NVIDIA paid for Run:ai — validating GPU orchestration software



Samsung building 50,000-GPU on-prem AI factory.
Private sector owns 80% of global AI compute (Epoch AI).

BUSINESS MODEL

Enterprise SaaS, per GPU

Software-only. Deployed on customer hardware. 80–90% gross margins.

PLATFORM LICENSE

~$100 /GPU/month

B2B SaaS · AI Infrastructure · 80-90% margins

REVENUE PROJECTIONS

1,000 GPUs	→ $100K/mo	$1.2M ARR
10,000 GPUs	→ $1M/mo	$12M ARR
100,000 GPUs	→ $10M/mo	$120M ARR

COMPETITIVE LANDSCAPE

No one owns on-prem training reliability

Cloud platforms handle this invisibly. On-prem teams piece together scripts that break.

	ON-PREM	TRAINING-AWARE	AUTO-RECOVERY	OBSERVABILITY
AWS SageMaker / HyperPod	✗	✓	✓	✓
Run:ai (NVIDIA)	✓	✗	✗	Partial
Slurm + DCGM + scripts	✓	✗	✗	✗
Sixtytwo	✓	✓	✓	✓

TEAM

Built by engineers who know this stack

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CO-FOUNDER

- Stanford CS '26
- SWE Intern, Meta — LLM content integrity
- Research Asst., NCSU — DNA data storage
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- TreeHacks 1st Place, OpenAI Prize

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- Stanford OVAL AI Lab — NLP research
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- Published at NeurIPS, MICCAI
- SWE Intern, Amazon — LLM guardrails
- Research Fellow, Hoover Institution

All technical founders · 3 years building together · Full-time committed

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The future of AI runs on-prem.

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